

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Year: 2024

Subject: 031021152 Electronic circuits II

Section: 05-06

Date: 9 January 2025

Time: 08:00 – 10:00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your competency on this subject. The test is divided into 3 sections. Here are 5 pages (including this page) and worth 35 points.

- Answer the questions on the answer book.
- Books, documents and lecture notes are not allowed.
- Student must be in the exam room for one hour once the exam is started and, while taking the exam.
- Students are not allowed to go out except in an emergency case.
- Before leaving exam room, students should make sure that they do not take this test paper outside the exam room.
- Students are not allowed to use any electronic communication device.
- Calculator are allowed to use in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is a part of set used to access students' abilities as follows:

CLO	Descriptions	Part A	Part B	Part C
1	Explain transistors, their types, depletion region, Bias in transistors, JFETs and types	✓	✓	✓
2	Identify and analyse FET basing, gate biasing, self-biasing and voltage divider biasing and with their transfer characteristics	✓	✓	✓
3	Identify and differentiate MOSFET and its types. Analyse types of E-MOSFET biasing and D-MOSFET biasing	✓	✓	✓
4	Explain amplifiers and cascaded amplifiers	✓	✓	✓

Formulae:

JFET Fixed-bias	$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$ $I_D = I_S, \quad I_G = 0A$ $V_{GSQ} = -V_{GG}, \quad V_{DS} = V_{DD} - V_{GG}$
JFET Self-bias	$V_{GS} = -I_D R_S$ $V_{DS} = V_{DD} - I_D (R_D + R_S)$
JFET (Gate bias)	$V_{GS} = V_{SS} - I_D R_S$ $V_{DS} = V_{DD} + V_{SS} - I_D (R_D + R_S)$
JFET (Voltage divider bias)	$V_G = \frac{R_2 * V_{DD}}{(R_1 + R_2)}$ $V_{GS} = V_G - I_D R_S$ $V_{DS} = V_{DD} - I_D (R_D + R_S)$
Depletion type MOSFET (Fixed bias)	$V_{GSQ} = +V_{GG}, \quad V_{DS} = V_{DD} - I_D R_D$
Depletion type MOSFET (Voltage divider bias)	$V_G = \frac{R_2 * V_{DD}}{(R_1 + R_2)}$ $V_{GS} = V_G - I_S R_S$ $V_{DS} = V_{DD} - I_D (R_D + R_S)$
Enhancement type MOSFET (Fixed bias)	$V_{GS} = V_{DS}$ $V_{GS} = V_{DD} - I_D R_D$
Enhancement type MOSFET (Voltage divider bias)	$V_G = \frac{R_2 * V_{DD}}{(R_1 + R_2)}$ $V_{GS} = V_G - I_D R_S$

V_{GS} versus I_D Using Shockley's Equation

V_{GS}	I_D
0	I_{DSS}
$0.3V_P$	$I_{DSS}/2$
$0.5V_P$	$I_{DSS}/4$
V_P	0 mA

⇒ Q1A1 = Question number 1 Part A Chapter 1

⇒ Q1B2 = Question number 1 Part B Chapter 2

⇒ Q1C3 = Question number 1 Part B Chapter 3

Part A: Write short answer for any ten questions at least 2 questions from each chapter.

(10X0.5)

5 Points

Q1A1. What is a transistor?

Q2A1. Write the name of three terminals of a transistor?

Q3A1. What is the region between P-N junction called in a transistor?

Q4A2. What are the three types JFET biasing?

Q5A2. Is current present in JFET biasing?

Q6A2. What is pinch of voltage in JFET?

Q7A3. What are two types of MOSFET?

Q8A3. Can I_{DSS} be more than drain current in MOSFET?

Q9A3. Is a channel always present in MOSFET?

Q10A4. Can the gate to source voltage in Depletion metal oxide semiconductor field effect transistor range between positive and negative voltage?

Q11A4. What is a Q-point in a MOSFET?

Q12A4. What is the threshold voltage in MOSFET?

Q13A5. What is an active device?

Q14A5. What is an ideal amplifier?

Q15A5. What are dependent sources?

Part B: Answer any five at least 1 question from each chapter. (5X2)

10 Points

Q1B1. Draw and explain basic operation of n-channel JFET.

Q2B1. Write the difference between BJT and FET?

Q3B2. Write shorthand method to plot transfer characteristics of JFET.

Q4B2. Draw gate bias JFET along with the formulae to calculate V_{GS} and I_D .

Q5B3. Sketch the transfer characteristics for DMOSFET with $I_{DSS} = 6$ mA and $V_P = 8$ V.

Q6B3. Why the transfer function cannot be defined by Shockley's equation in EMOSFET?

Q7B4. What is the main difference between JFET and depletion-type MOSFET biasing?

Q8B4. Draw feedback biasing enhancement type MOSFET and write the formulae to calculate gate to source voltage along with drain current.

Q9B5. What are the four fundamental types of amplifying structures in active devices?

Q10B5. Draw and label realistic amplifier along with formula to calculate V_{in} .

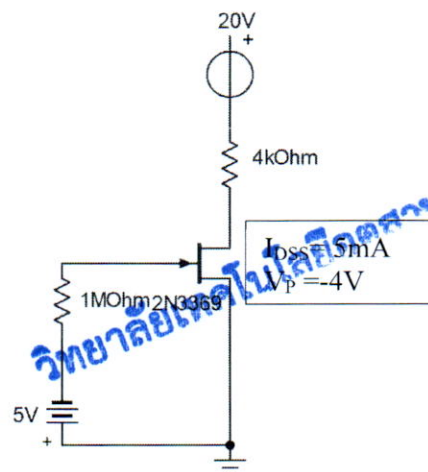
Part C: Solve the following questions. (5X4)

20 Points

Q1C1. From the JFET data sheet, it indicates that $I_{DSS} = 9\text{mA}$ and $V_{GS(off)} = -8\text{V}$ (maximum).

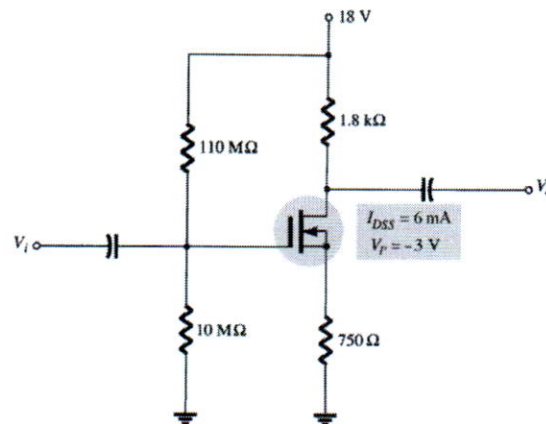
Determine the drain current for $V_{GS} = 0\text{V}$, -1V and -4V and sketch the I_D vs V_{GS} characteristic curve (Don't use short method).

Q2C2. Determine V_{GSQ} , I_{DQ} , V_{DS} , V_D , V_G , V_S from the following circuit using graphical method.

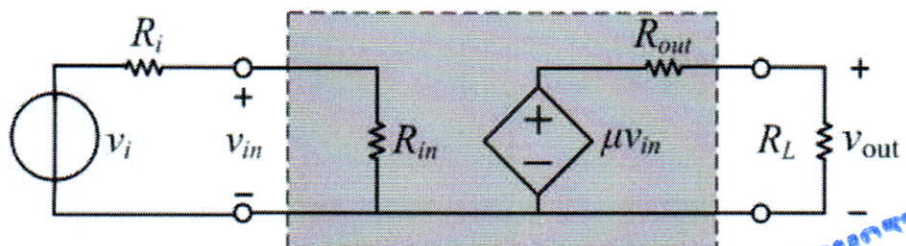


Q3C3. Given $V_{GS(Th)} = 4\text{V}$ and $I_{D(on)} = 4\text{mA}$ at $V_{GS(on)} = 6\text{V}$, determine k and write the general expression for I_D . Sketch the transfer characteristics for the device. Determine I_D for the device at $V_{GS} = 2\text{V}$, 5V , and 10V .

Q4C4. For the n-channel depletion-type MOSFET, determine: I_{DQ} , V_{GSQ} and V_{DS} .



Q5C5. Determine the voltage, current and power gains.



GOOD LUCK!

วิทยาลัยเทคโนโลยีคณาจารย์

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